

NOTE:

- Create a folder to save your works. Name the folder appropriately (e.g. Lab 2)
- Use comment `//` to write your name, ID, Group and Lab Question in each program.
- Save your file as `.cpp` (programming) and word document (diagram).
- Submission depending on your lecturer's instruction

LAB 1 : INTRODUCTION TO DATA STRUCTURE AND ALGORITHMS.

1. Write a program that prints the maximum of four given integers. Use the *cin* input operator to insert the four numbers entered by user and use the *cout* output operator to display the maximum number.

Sample output:

```
Enter four integers : 44  33  66  55
Their maximum is 66
```

2. Write a program that will tell the user if the number that he/she entered is even or odd. The program will exit if user enter 0. Create two functions *void even* and *void odd*.

Sample output:

```
Type a number (0 to exit): 9
Number is odd.
Type a number (0 to exit): 6
Number is even.
Type a number (0 to exit): 0
Thank you.
```

3. Write a program to imitate the calculator. Use *switch case* statement.

Sample Output :

```
Enter an operator (+,-,*,/) : +
Enter two operands : 10 20
10 + 20 = 30
```

4. Change the following code from *for* loop to *while* loop.

```
#include <iostream.h>
int main()
{
    for(int counter=1; counter<=10; counter++)
        cout<<counter<<endl;

    return 0;
}
```

5. A class of five students took a quiz. Write a C++ code statement that will prompt the user to enter five student's marks to calculate the average quiz marks. [Hint: Use while loops to print all the five marks enter by user.]

Sample output:

```
Enter mark: 56
Enter mark: 88
Enter mark: 95
Enter mark: 45
Enter mark: 78
Mark average is: 72
Press any key to continue
```

6. Create an array that prompt user to key in five integer values. Pass the array to a function name *Sum()* that will calculate and display the sum of all the elements.

Sample output:

```
Enter five numbers and the program will calculate sum
of the numbers.
Enter number 1: 1
Enter number 2: 6
Enter number 3: 4
Enter number 4: 3
Enter number 5: 3
You have entered : 1, 6, 4, 3, 3
Sum of the numbers entered is : 17
```

7. The following program is written to get a software details from the user.
Write a code to complete the program based on the instructions given.

```
#include<iostream>
#include<string.h>
using namespace std;

class software
{
    string name; int id,qty; float price,total;
public:
    ..... (i) .....
    {
        .....
    }

    ..... (ii) .....
    {
        .....
    }

    ..... (iii) .....
    {
        .....
    }

    ..... (iv) .....
    {
        .....
    }

};

int main()
{
    software s;
    string name;
    int id,qty;
    float price;

    cout<<"Enter software name :";
    getline(cin,name);
    cout<<"Enter software quantity :"; cin>>qty;
    cout<<"Enter software ID :"; cin>>id;
    cout<<"Enter software price :"; cin>>price;
    s.setdata(name,id,qty,price);
    s.calculate_total();
}
```

```
s.display();
}
```

Sample Output

```
A list of computer software.
Enter software name :Microsoft Word
Enter software quantity :4
Enter software ID :112211
Enter software price :345.00
Software Name           :Microsoft Word
Software Quantity       :4
Software ID             :112211
Software Price          :RM 345
Total                   :RM 1380
-----
```

- i. Write a function called *setdata(...)* that has four parameters and initialize the parameters to *name*, *id*, *qty* and *price*.
- ii. Write a function called *calculate_total()* that calculate the *total* price.
- iii. Write a function called *display()* that display all information as shown in the sample output.
- iv. Write a constructor for class *software* that displays a welcome note “A list of computer software.”